

High Altitude Long Endurance (HALE) Unmanned Aerial Vehicles (UAVs)

Sustained Flight through Renewable Energy

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Objective of the Project Work

Primary Objective:

Theoretical design and validation of a HALE Solar-Powered UAV for sustained flight.

Secondary Objectives of Project:

01

Conduct comprehensive literature review on solar-powered UAVs.

02

Identify limitations and propose theoretical improvements.

03

Use simulations for performance validation.

04

Analyze results and suggest future research directions.

Literature Survey

establishing context, identifying research gaps, and validating a study's relevance by summarizing existing knowledge

01

Solar-Powered UAVs:

Reviewed key papers and studies on solar-powered UAV technologies. Researched on solar cells used in UAVs.

Solar cell technologies: triple-junction cells, thin-film solar cells.

02

Energy Storage Systems:

Explain the various energy storage systems used in UAVs, including batteries and fuel cells.

Energy storage systems: lithium-ion batteries, fuel cells.

Literature Survey

establishing context, identifying research gaps, and validating a study's relevance by summarizing existing knowledge

03

Solar Cell Efficiency:

An analysis of the efficiency of current solar-powered UAV designs was conducted.

Triple-junction solar cells: ~25–30% efficiency.

Thin-film cells: lower efficiency but lighter weight.

04

Energy Storage Efficiency:

Explored the diverse energy storage systems employed in Unmanned Aerial Vehicles (UAVs), encompassing both batteries and fuel cells.

Batteries: high energy density, but degrade over cycles.

Fuel cells: stable efficiency, but add system complexity.

Literature Survey

establishing context, identifying research gaps, and validating a study's relevance by summarizing existing knowledge

05

Energy Storage:

Identified gaps in energy storage and conversion efficiency.

Night-time endurance limited by current batteries.

Conversion losses in storage cycles.

06

Design Limitations:

Highlighting limitations in current UAV designs, such as weight constraints and aerodynamic efficiency.

Weight constraints impact aerodynamic efficiency.

Limited focus on hybrid energy systems in past research.

Resources Required



Database

Academic Database which supporting the research objectives, Technical Papers, PDRs

Tools like MATLAB, Simulink for data analysis or simulations.
Ansys, CATIA for modelling

Simulation Software

Expert Guidances

Consultation with faculty and aerospace engineers.

Project roadmap

	Month 01	Month 02
	Brainstorming on research domain, summarizing key design principles and technologies.	Preliminary design, theoretical framework
Month 03	Month 04	Goal
Results & analysis	Final documentation & recommendations	1. Theoretical framework for HALE UAV design. 2. Identification of efficiency improvements. 3. Simulation-based validation of feasibility. 4. Foundation for future experimental research.



Any questions?
Ask away!

Let's discuss any doubts or concerns.
